Radar range and velocity

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Separate a radar's range measurement from its line-of-sight velocity measurement.

SUCCESS CRITERIA - I can...

1. Use round-trip time for range.
2. Explain the information in a Doppler shift.
3. State what radial velocity leaves unknown.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use "round trip" and "signal" to explain your model or result. Sentence starters are optional.

round trip: travel from start to a reflector and back

signal: a disturbance that carries information

Doppler shift: a frequency change associated with relative motion

01 Bring back an earlier idea

An EM wave in vacuum has frequency 5×10^{9} Hz.
Use $c=3.00\times10^{8}$ m/s.
Find wavelength and name the spectral region.

Word help: vacuum = space with negligible matter

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: range = $c\Delta t/2$; $c = 3.0\times10^{8}$ m/s.
Radar time is a round trip. Doppler gives line-of-sight velocity information.

Worked reasoning: A radar echo returns after 2 microseconds. With $c=3\times10^{8}$ m/s, range$=c\Delta t/2=300$ m. The measured time includes the outward and return paths.

Your turn: Draw the round trip and label the conversion 2 microseconds$=2\times10^{-6}$ s.

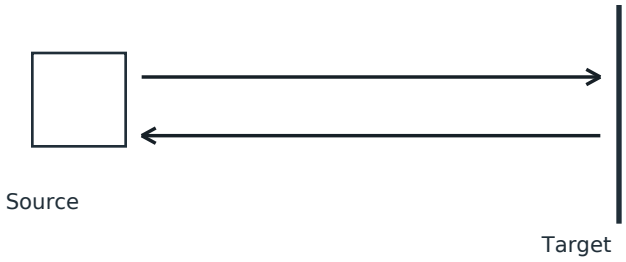
Sentence starter: The ... in my model represents ... . Its label shows ... .

Marzano check - circle: ① ② ③

03 Represent and interpret

Draw outgoing and returning radar paths. Which measurement helps range: echo delay or frequency shift? Which helps radial velocity?

Sentence starter: The ... in my model represents ... . Its label shows ... .



Marzano check - circle: ① ② ③

04 Solve and justify

A radar echo returns after .040 s.
Find range. A monostatic radar has$\lambda=.10$ m and shift magnitude 400 Hz; use$|v_r|=|\Delta f|\lambda/2$ to find radial speed.

Word help: magnitude = size of a quantity without its direction

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Analyze the reasoning

An asteroid has zero measured radial velocity at one instant. Does that prove it is stationary?
Sketch another possible motion.

Sentence starter: The ... in my model represents ... . Its label shows ... .

Marzano check - circle: 1 2 3

06 Choose a method

A wave travels 12 m in 0.4 s at frequency 60 Hz.
Find speed and wavelength.

Word help: wavelength = distance between corresponding points on repeated waves

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: 1 2 3

07 Exit check

Two targets at the same range have different Doppler shifts.
Explain why they can have same time-delay but different motions.

Sentence starter: Both ... . They differ in ... because ... .

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

Radar range changes from 1500 to 1440 m over 0.5 s.
Find average radial velocity taking outward positive.
Explain why total speed may be larger.

Sentence starter: Both ... . They differ in ... because ... .

Marzano check - circle: 1 2 3